NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

CCDS25029 - Can Deep Neural Networks Discover Chess Geometry Without Knowledge of Game Rules?

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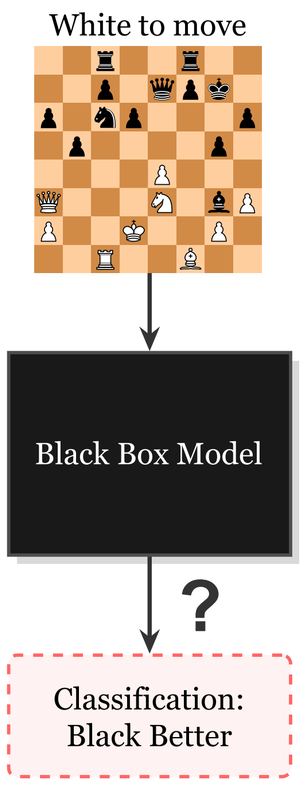
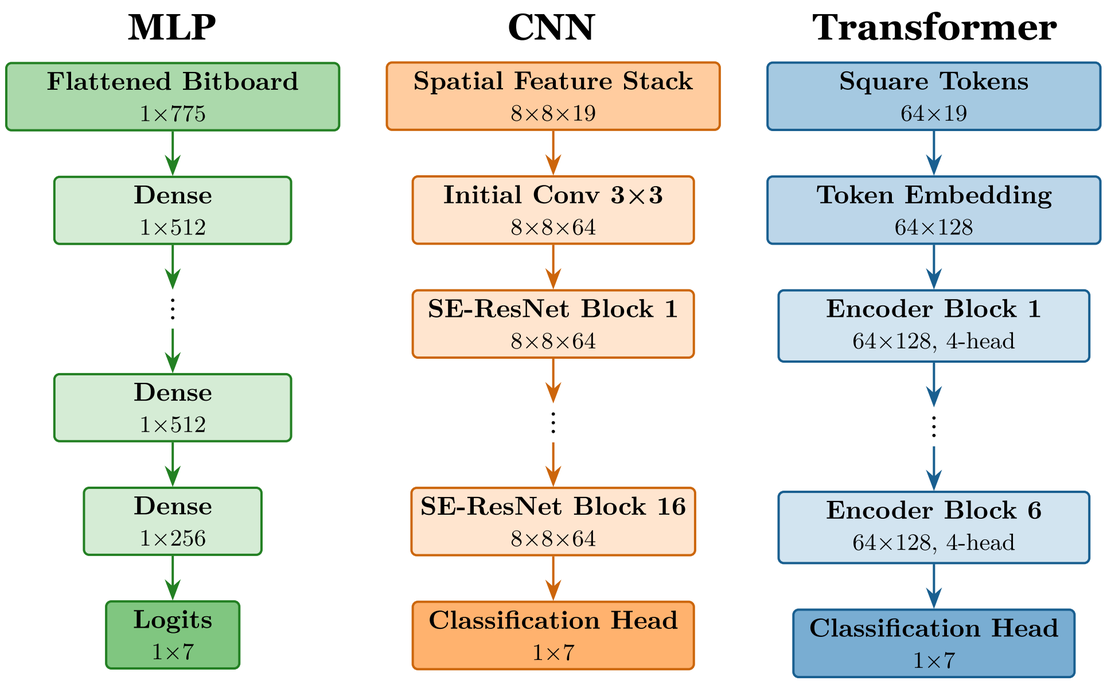
Introduction



- Explainable AI is a crucial field of research, with chess often serving as a testbed with its perfect-information environment.
- While domain-specific knowledge and deep search algorithms enable superhuman play, their complexity obscures a network's true internal representations.
- In this study, we investigate how Multi-Layer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), and Transformers internalize the geometry of chess without rules, context, or search.
- By visualizing and probing these internal representations, we reveal how distinct inductive biases dictate the scale of their spatial understanding.

Methodology

We use a supervised learning approach. Models are trained purely on a numerical encoding of static board states to predict a 7-class position evaluation generated by Stockfish 17 at depth 20. Gaussian label smoothing is applied to soften the rigid 7-class boundaries.



Dataset
5.8M unique positions, augmented with random legal moves to introduce more imbalanced states, then class balanced for training.

Positional Accuracy (Evaluation Metric)
We coin this metric to measure model performance on states where material is perfectly equal, but the position evaluation is decisive.

Mechanistic Interpretability
Linear probes are trained on frozen intermediate activations of each model to predict legal piece movements.

Saliency Maps
Gradients are summed across all piece types (i) to determine which specific board squares (j) dictate the model's decision.

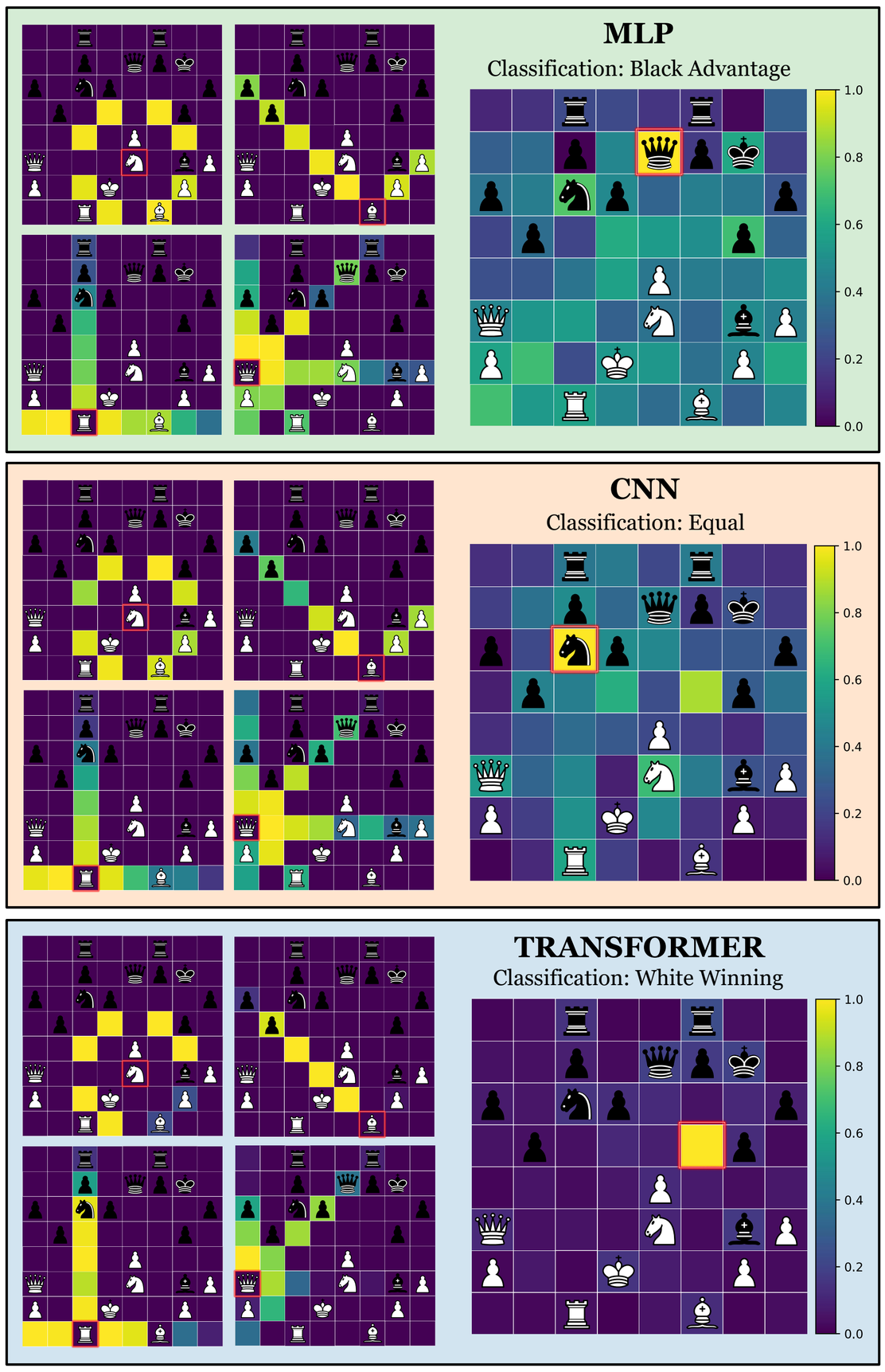
$$S_j = \sum_{i=1}^{12} \left| \frac{\partial y_c}{\partial x_{i,j}} \right|$$

Results

Performance comparison

Architecture	Params	Top-1 Acc.	Off-by-1 Acc.	MAE	Positional Acc.
MLP	1.32M	58.18%	88.05%	0.622	56.66%
CNN	1.22M	69.87%	94.72%	0.384	70.08%
Transformer	1.21M	**73.04%**	**95.83%**	**0.333**	**73.21%**

Legal Move Probe Maps **Saliency Map**



The Transformer's self-attention captures global piece interactions far better than localized CNN convolutions or MLPs, which rely on raw capacity and default to simple piece counting.

Conclusion

We open these black boxes and showed that they can internalize complex board geometry entirely from static states. However, overcoming the absence of explicit rules or search relies heavily on selecting an architecture with an optimal spatial inductive bias.